

```
import os
```

```
!wget https://dosairnowdata.org/dos/historical/Colombo/2018/Colombo_PM2.5_2018_YTD.csv -P ./data/
```

```
--2023-11-12 04:18:19-- https://dosairnowdata.org/dos/historical/Colombo/2018/Colombo_PM2.5_2018_YTD.csv
Resolving dosairnowdata.org (dosairnowdata.org)... 74.208.236.6, 2607:f1c0:100f:f000::279
Connecting to dosairnowdata.org (dosairnowdata.org)|74.208.236.6|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 852964 (833K) [text/csv]
Saving to: './data/Colombo_PM2.5_2018_YTD.csv'
```

```
Colombo_PM2.5_2018_ 100%[=====>] 832.97K 847KB/s in 1.0s
```

```
2023-11-12 04:18:21 (847 KB/s) - './data/Colombo_PM2.5_2018_YTD.csv' saved [852964/852964]
```

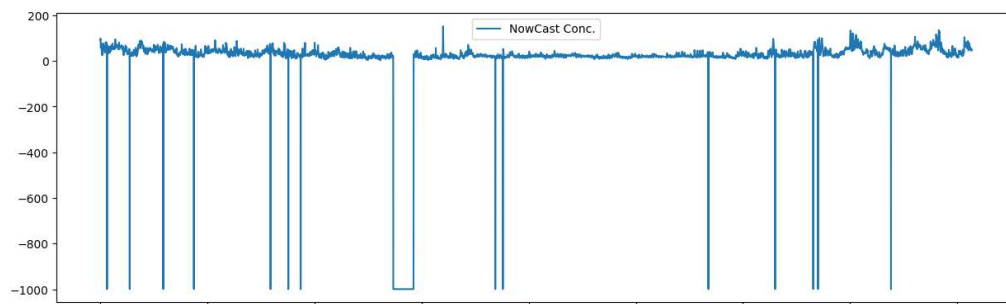
```
os.listdir('./data/')
```

```
import pandas as pd
df = pd.read_csv('./data/Colombo_PM2.5_2018_YTD.csv')
```

```
df.head()
```

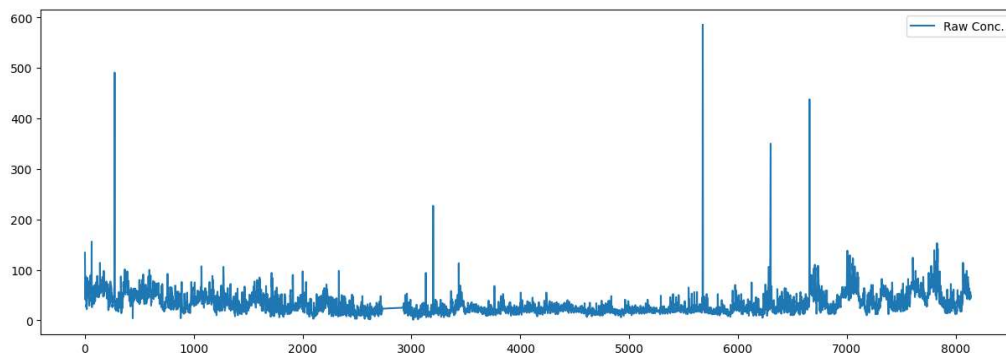
	Site	Parameter	Date (LT)	Year	Month	Day	Hour	NowCast Conc.	AQI	AQI Category	Raw Conc.	Con Un
0	Colombo	PM2.5 - Principal	2018-01-01 01:00 AM	2018	1	1	1	92.5	170	Unhealthy	134.0	UG/I
1	Colombo	PM2.5 - Principal	2018-01-01 02:00 AM	2018	1	1	2	97.8	173	Unhealthy	103.0	UG/I
2	Colombo	PM2.5 - Principal	2018-01-01 03:00	2018	1	1	3	89.9	169	Unhealthy	82.0	UG/I

```
df[['NowCast Conc.', 'Raw Conc.']].plot(subplots=True, figsize=(15,10));
```



```
df[df['Raw Conc.'] > 0]['Raw Conc.'].plot(figsize=(15,5), legend=True)
```

<Axes: >



```
df['Date (LT)'] = pd.to_datetime(df['Date (LT)'])
df.set_index('Date (LT)', inplace=True)
df.head()
```

	Site	Parameter	Year	Month	Day	Hour	NowCast Conc.	AQI	AQI Category	Raw Conc.	Con Un
Date (LT)											
2018-01-01 01:00:00	Colombo	PM2.5 - Principal	2018	1	1	1	92.5	170	Unhealthy	134.0	UG/I
2018-01-01 02:00:00	Colombo	PM2.5 - Principal	2018	1	1	2	97.8	173	Unhealthy	103.0	UG/I
2018-01-01 03:00:00	Colombo	PM2.5 - Principal	2018	1	1	3	89.9	169	Unhealthy	82.0	UG/I

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 8140 entries, 2018-01-01 01:00:00 to 2019-01-01 00:00:00
Data columns (total 13 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Site         8140 non-null   object
1   Parameter    8140 non-null   object
2   Year         8140 non-null   int64
3   Month        8140 non-null   int64
```

```

4   Day                8140 non-null   int64
5   Hour               8140 non-null   int64
6   NowCast Conc.     8140 non-null   float64
7   AQI               8140 non-null   int64
8   AQI Category      7929 non-null   object
9   Raw Conc.         8140 non-null   float64
10  Conc. Unit        8140 non-null   object
11  Duration          8140 non-null   object
12  QC Name           8140 non-null   object
dtypes: float64(2), int64(5), object(6)
memory usage: 890.3+ KB

```

```

df.drop(columns=['Site', 'Parameter', 'Year', 'Month', 'Day', 'Hour', 'Conc. Unit', 'Duration'], inplace=True)
df.head()

```

	NowCast Conc.	AQI	AQI Category	Raw Conc.	QC Name
Date (LT)					
2018-01-01 01:00:00	92.5	170	Unhealthy	134.0	Valid
2018-01-01 02:00:00	97.8	173	Unhealthy	103.0	Valid
2018-01-01 03:00:00	89.9	169	Unhealthy	82.0	Valid
2018-01-01 04:00:00	72.9	160	Unhealthy	56.0	Valid
2018-01-01 05:00:00	56.9	152	Unhealthy	41.0	Valid

```

df2 = df[df['Raw Conc.']>=0]
df2.describe()

```



	NowCast Conc.	AQI	Raw Conc.
count	7899.000000	7899.000000	7899.000000
mean	29.241853	88.832257	32.177744
std	54.750358	64.285820	22.010910
min	-999.000000	-999.000000	0.000000
25%	19.300000	66.000000	19.000000
50%	26.100000	80.000000	26.000000
75%	40.000000	112.000000	40.000000
max	151.800000	202.000000	586.000000

```
df2.info()
```

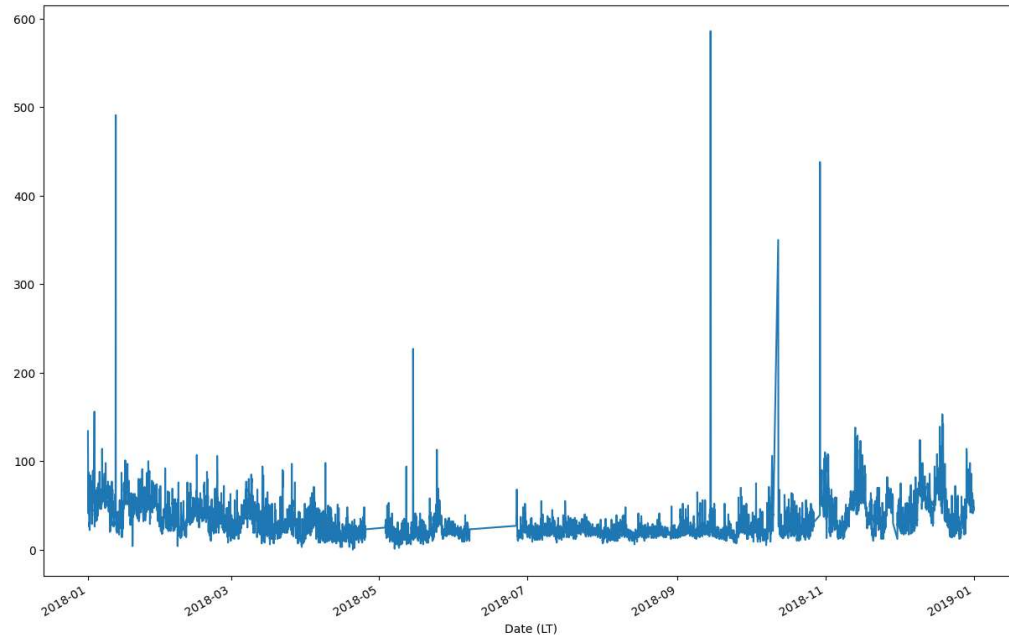
```

<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 7899 entries, 2018-01-01 01:00:00 to 2019-01-01 00:00:00
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   NowCast Conc.   7899 non-null   float64
1   AQI             7899 non-null   int64
2   AQI Category    7879 non-null   object
3   Raw Conc.       7899 non-null   float64
4   QC Name         7899 non-null   object
dtypes: float64(2), int64(1), object(2)
memory usage: 370.3+ KB

```

```
df2['Raw Conc.'].plot(figsize=(15,10))
```

<Axes: xlabel='Date (LT) '>



```
df2[['Raw Conc.']].resample('1D').mean().plot(figsize=(15,10) , kind='line')
```

<Axes: xlabel='Date (LT)'\>

